

Mohammad Savari, EIT

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Professional Summary

Electrical and systems engineer with experience in system modeling, control systems, sensor calibration, sensor fusion, embedded systems, PLC platforms, field testing, technical documentation, and data-driven analysis. Ph.D. candidate in Mechanical Engineering at the University of Ottawa with research expertise in system dynamics, complex systems, computational modeling, multi-agent systems, and reinforcement learning. Skilled in Python, MATLAB, R, LabVIEW, Arduino, Atmel microcontrollers, LOGO, and Siemens Step 7.

Technical Skills

Control and Systems: Control systems, system dynamics, system modeling, simulation, performance optimization, design validation, troubleshooting

Embedded and Automation: Arduino, Atmel microcontrollers, PLC platforms, LOGO, Siemens Step 7, embedded systems, automation systems

Sensors and Testing: Sensor calibration, sensor fusion, data validation, field testing, system performance evaluation, reliability improvement

Programming and Analysis: Python, MATLAB, R, LabVIEW, statistical analysis, data processing, computational modeling

Research and Advanced Methods: Multi-agent systems, reinforcement learning, complex systems modeling, network development, simulation-based research

Education

Ph.D. in Mechanical Engineering

University of Ottawa

2021–2026 expected

Ottawa, Canada

- Thesis: *Fostering Consensus Using Social Contagion.*
- Research focus: complex systems, system dynamics, computational modeling, network science, and consensus formation.
- Awarded full admission scholarship and merit scholarship from the University of Ottawa.

Bachelor of Science in Electrical Engineering

2014–2018

Engineering Experience

System Engineering Intern

AirShare Inc.

Sep 2023–Mar 2024

Ottawa, Canada

- Developed and implemented system models to support performance optimization and design validation.
- Established and maintained communication frameworks to improve system reliability and operational stability.
- Supported sensor calibration, sensor fusion, and data validation activities to improve system-level accuracy.
- Coordinated procurement and field testing activities to support efficient project execution and high-quality outcomes.
- Assisted with troubleshooting, technical documentation, and system performance evaluation.

Project Manager

Centre for Entrepreneurship and Engineering Design

Sep 2024–Present

Ottawa, Canada

- Manage four interdisciplinary engineering teams developing accessibility-focused devices from concept to prototype.
- Review technical documentation, design reports, and project deliverables to ensure alignment with client requirements.
- Monitor design progress, project timelines, milestones, and risk areas to support successful execution.
- Coordinate procurement of components, materials, and resources required for prototype development.
- Provide guidance on feasibility, usability, documentation quality, and client-centered engineering design.

Teaching Assistant

University of Ottawa

Jan 2022–Present

Ottawa, Canada

- Supported courses related to Mechatronics, Systems Dynamics, Biomedical Systems Dynamics, Machine Learning in Control, Robotics, and Product Development.
- Assisted students with system modeling, control concepts, robotics, mechatronics, and machine learning for control.
- Provided feedback on assignments, projects, and technical submissions to improve engineering learning outcomes.

Research Experience

Social Contagion Researcher

Faculty of Engineering, University of Ottawa

Sep 2021–Present

Ottawa, Canada

- Develop computational models to study consensus formation and information propagation in dynamic networked systems.
- Integrate system dynamics concepts into the analysis of social contagion processes on complex networks.
- Develop heterogeneous scale-free network models with tunable clustering properties.
- Implement distribution likelihood analyses to evaluate scale-free characteristics in generated network models.

Complex Networks Researcher

Telfer School of Management, University of Ottawa

May 2025–Present

Ottawa, Canada

- Analyze sub-networks, communities, and topological properties in complex global networks.
- Collect, clean, and integrate data using web scraping, natural language processing, manual extraction, and entity matching.
- Generate network visualizations and computational models using Python and Gephi.

Additional Professional Experience

Search Engine Evaluator

TELUS International

Jun 2024–Present

- Evaluate search engine results for relevance, accuracy, completeness, and alignment with user intent.
- Review online content for quality, reliability, usability, and cultural appropriateness.
- Apply structured evaluation guidelines with consistency, attention to detail, and analytical judgment.

Front-End Associate

Walmart Canada

May 2022–Sep 2024

Ottawa, Canada

- Delivered customer service in a high-volume retail environment and supported efficient checkout operations.
- Processed transactions accurately, handled payments, and resolved customer inquiries.

Selected Publications and Presentations

- **Savari, M.**, et al. 2025. “How Social Contagions Shape Collective Consensus in the Presence of Scale-Free Networks.” Submitted.
- **Savari, M.**, et al. 2025. “Leveraging Social Contagion to Foster Consensus in Collective Decision-Making.” International Conference on Computational Science, Singapore.